

Tripti Gupta

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PROFILE SUMMARY

AI/ML Engineer with hands-on experience building LLM-based systems, automation workflows, and computer vision applications. Worked on locally deployable LLM pipelines, UI element detection using YOLO, and integrating AI models into real-world automation use cases. Skilled in Python, prompt engineering, and designing end-to-end AI solutions. Currently exploring agentic AI concepts using frameworks like LangChain and LangGraph.

EXPERIENCE

- Watermelon Software Inc.** Oct 2025 – Present
AI Engineer Remote
 - Worked on a computer vision-driven automation solution involving UI element detection from application screens, where the trained model was integrated as a Chrome extension to enable intelligent interaction and improve the robustness of automated testing workflows.
 - Currently developing and designing an LLM fine-tuning and evaluation framework to transform large language models into smaller, efficient, and locally deployable models, aiming to reduce reliance on third-party AI providers, improve performance, and ensure data privacy.
- GLA University** June 2024 – July 2024
AI Trainee Remote
 - Built an AI-powered health assistant using LLM APIs for chatbot interaction and calorie estimation.
 - Integrated external AI APIs and handled structured user input/output pipelines.
 - Worked on prompt design and response evaluation.

EDUCATION

- GLA UNIVERSITY** 2022 - 2026
B.Tech in Computer Science (AI & ML) Mathura, India
- British English School** 2022
Intermediate Gaya, India
- Param Gyan Niketan** 2020
High School Gaya, India

PROJECTS

- TaskForge AI: Multi-Agent Workflow Automation System**
Techstack: CrewAI, Python, LLM APIs, Tools/API Integration
 - Built a multi-agent system using CrewAI where specialized agents collaboratively plan, execute, and validate complex tasks.
 - Implemented task decomposition and agent-based coordination to enable structured multi-step workflow execution.
 - Integrated external tools and APIs to allow agents to perform real-time actions and improve task reliability.
- DocuMind AI: Retrieval-Augmented Knowledge Intelligence System**
Techstack: DSPy, Transformers, Vector Retrieval
 - Developed an LLM-powered RAG system for context-aware question answering over custom document collections.
 - Built a document ingestion pipeline with chunking and embedding-based retrieval for efficient knowledge access.
 - Designed a modular retrieval-generation architecture using DSPy and improved response relevance through prompt and retrieval optimisation.
- FlowOps AI: Agentic Workflow Automation System**
Techstack: n8n, LLM APIs, Python, APIs
 - Built AI-driven workflows using n8n where agents dynamically process inputs and execute multi-step automation tasks.
 - Integrated LLM APIs with external services to enable intelligent decision-making and real-time task execution.
 - Designed event-driven pipelines with conditional logic to automate end-to-end workflows across multiple systems.

SKILLS

- **Programming Languages:** Python , SQL
- **Database Systems:** MySQL, PostgreSQL
- **Data Science & Machine Learning:** Pandas, NumPy, Scikit-learn, Data Preprocessing Pipelines, Model Training & Evaluation
- **AI/Deep Learning:** PyTorch, TensorFlow, HuggingFace Transformers LLMs, RAG Pipelines, Prompt Engineering
- **Computer Vision:** YOLOv8, DeepSORT, Mediapipe, OpenCV
- **Agentic Frameworks:** LangChain, LangGraph (basic), CrewAI (learning)
- **Workflow Automation:** n8n

ACHIEVEMENTS

- 1st Place — Technical Poster Presentation (IBM).
- Certification — Business Intelligence using Power BI (Skill Nation).
- Certified — Intermediate SQL (HackerRank).
- Top 20 — IBM Ice Day Solvathon.
- Advisor — NSS Society, GLA University.